Congruence Compression and Simultaneous Hilbert Irreducibility

With an independent application to Baker's question

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Abstract

Let K be a number field and let $T \subset K$ be thin. We prove that its union of affine transforms $a(z)T + h(z)$, with z in a fixed finitely generated subgroup of a torus, has lattice density zero among algebraic integers when a is a nonzero monomial and h is a Laurent polynomial. In fixed integral-basis boxes of radius H, the exceptional count is $O(H^{[K:\mathbb{Q}]} \exp\{-c \log H/\log\log H\})$. A separate congruence argument gives the corresponding one-dimensional bound on the rational-integer slice. The proof combines a prime-compression consequence of Pollack's argument with a local deficit for each individual curve cover; the cover index is kept fixed before the Chinese remainder theorem is applied. This yields irreducibility of finitely many prescribed polynomial specializations simultaneously under all allowed affine transforms, for a density-one set of integral parameters. An elementary finite-field value-set lemma also treats parameter-dependent polynomials of every degree at least two whose leading coefficient is a monomial. The classical inputs, the relation to an earlier square-value result, and the limits of the conclusions are stated explicitly. An independent appendix proves rational linear independence of Dirichlet L-values at 1 for a specified infinite family of moduli by a separate cyclotomic argument.

Keywords. Thin sets; Hilbert irreducibility; S-units; finite-field value sets; congruence sieve; algebraic integers.

1 Statements and conventions

The purpose of this paper is to separate two sources of complexity in a uniform specialization problem. A thin set has a local deficit at suitably chosen primes, but an unbounded collection of multiplicative parameters seems to introduce too many translates of that set. At a carefully chosen product of primes, all parameters reduce through one small finite quotient. The local deficits then dominate the number of parameter classes.

Fix a number field K of degree $D = [K : \mathbb{Q}]$ and an integral basis $\boldsymbol{\omega} = (\omega_1, \dots, \omega_D)$ of $\mathcal{O}_K$. For $H \geq 1$, put

$$\mathcal{B}_K(H) = \left\{ \sum_{j=1}^{D} n_j \omega_j : n_j \in \mathbb{Z},\ |n_j| \leq H \right\}.$$

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Density below means density in these boxes. The conclusions hold for every fixed choice of integral basis. All constants may depend on the fixed arithmetic and geometric data. No uniformity in the degrees, field, generators, or coefficients is implicit.

A Laurent polynomial in $z = (z_1, \dots, z_k)$ belongs to $K[z_1^{\pm 1}, \dots, z_k^{\pm 1}]$. A nonzero monomial means

$$a(z) = cz_1^{\alpha_1} \cdots z_k^{\alpha_k}, \qquad c \in K^\times, \quad \alpha_j \in \mathbb{Z}.$$

Let $\Gamma \leq (K^\times)^k$ be finitely generated, with rank r and finite torsion subgroup of order w. For $N \geq 1$, its subgroup of Nth powers is denoted by Γ^N; in particular,

$$|\Gamma/\Gamma^N| \leq wN^r. \tag{1}$$

Definition 1.1. A subset $T \subset K$ is *thin* if it is contained in a finite union of a finite set and sets

$$f(C(K)) \cap K,$$

where C is a smooth projective geometrically integral curve over K and $f : C \to \mathbb{P}^1_K$ is a morphism of degree at least two.

This is the usual arithmetic notion of thinness on the affine line. Normalization and projective completion give the displayed form from the standard definition by generically finite covers, with finitely many exceptional images absorbed into the finite set; see [3, Chapter 3]. It is enough that T be a subset of such a union.

Theorem 1.2 (Affine transforms of a thin set)**.** *Let $T \subset K$ be thin, let a be a nonzero monomial, and let h be a Laurent polynomial. Define*

$$\mathcal{R} = \mathcal{O}_K \cap \{a(z)y + h(z) : y \in T,\ z \in \Gamma\}. \tag{2}$$

Then the following assertions hold.

(i) *For every $\varepsilon > 0$ there is a squarefree positive rational integer m such that*

$$|\mathcal{R} \bmod m\mathcal{O}_K| \leq \varepsilon m^D.$$

(ii) *There are $c > 0$ and H_0 such that, for $H \geq H_0$,*

$$|\mathcal{R} \cap \mathcal{B}_K(H)| \ll H^D \exp\left\{-c\frac{\log H}{\log\log H}\right\}.$$

In particular, $\mathcal{R}$ has density zero.

(iii) *There is also a constant $c' > 0$ such that*

$$|\mathcal{R} \cap \mathbb{Z} \cap [-H, H]| \ll H \exp\left\{-c'\frac{\log H}{\log\log H}\right\}.$$

The last assertion concerns the rational-integer slice and is not inferred from the ambient density statement.

The theorem asserts density zero, not that the union in (2) is itself thin. It allows arbitrarily large positive or negative exponents in Γ and arbitrary cancellation in $h(z)$. The number of multiplicative coordinates and the Laurent polynomial are fixed throughout.

Theorem 1.3 (Simultaneous Hilbert irreducibility)**.** *Let $P_1(T,Y),\dots,P_s(T,Y)\in K(T)[Y]$ be irreducible, of positive degree in Y, and let G_i be the generic splitting-field Galois group of P_i over $K(T)$, with its action on the roots. Fix a,h,Γ as above. Outside a set of $\mathcal{O}_K$ satisfying the bounds in Theorem 1.2(ii)–(iii), every $t\in\mathcal{O}_K$ has the following property: for every $1\le i\le s$ and every $z\in\Gamma$,*

$$P_i\big(a(z)t+h(z),Y\big)$$

is defined, has its generic degree, is separable and irreducible over K, and has Galois group G_i with the generic permutation action, up to conjugacy.

The exceptional set controls the existential quantifier "there is a bad z". Its complement therefore controls *all* z simultaneously. This is different from a separate density statement for each fixed translate. A finite collection of distinct pairs (a,h) can also be allowed, by taking a finite union of exceptional sets.

We will also prove a result that is not merely an application of a fixed thin set, because its lower coefficients may vary with z.

Theorem 1.4 (Polynomial families of arbitrary degree)**.** *Suppose $d\ge 2$ and*

$$F(X,z)=cz^{\boldsymbol{\alpha}}X^d+\sum_{i=0}^{d-1}A_i(z)X^i,\qquad A_i\in K[z_1^{\pm1},\dots,z_k^{\pm1}],\quad c\neq 0.$$

The set

$$\mathcal{R}_F=\{t\in\mathcal{O}_K : t=F(x,z)\text{ for some }x\in K,\ z\in\Gamma\}$$

satisfies all three conclusions of Theorem 1.2. There is no parity restriction on d.

Relation to earlier work

The use of moduli with small multiplicative exponent is classical. Pollack [1] applies such moduli to sums from finitely generated multiplicative groups over number fields. We use a stronger quantitative form visible in his proof, not a new theorem about primes. The other input is the standard local deficit for an individual thin cover; a reference is Browning–Loughran [2, Lemma 3.8(2)]. We give a direct curve proof suited to the present application.

The transfer established here retains a free point on a curve while allowing all multiplicative parameter values. An earlier corpus manuscript [5] treats an even-degree polynomial family over $\mathbb{Q}$ and answers the square-value question raised after Example 1.3 of Corvaja–Zannier [4]. The present paper does not count that original question as newly solved again. Its exact scope and the absence of a historical priority claim are discussed in Section 8.

2 Primes with a compressed common exponent

We isolate the prime-number input. Relative density refers to density among rational primes.

Proposition 2.1 (External sieve input)**.** *For every $\delta>0$ there is $\eta\in(0,1)$ such that, for all sufficiently large X, the number of primes $p\le X$ for which $p-1$ has a prime factor greater than $X^{1-\eta}$ is less than $\delta X/\log X$.*

This is Pollack's Lemma 2.1 [1], a consequence of the Brun-sieve argument of Erdős. It is the only sieve estimate assumed here. We also use the prime number theorem and the Chebotarev density theorem.

Lemma 2.2 (Prime compression)**.** *If $\mathcal{P}$ has positive lower relative density, there are constants $\delta, \eta > 0$ such that, for every sufficiently large X, a finite $\mathcal{L} \subset \mathcal{P} \cap [\delta X/8, X]$ satisfies*

$$L := |\mathcal{L}| \geq \frac{\delta X}{4 \log X}, \tag{3}$$

$$\log N \leq X^{1-\eta} \log X, \qquad N := \operatorname{lcm}_{p \in \mathcal{L}}(p-1), \tag{4}$$

$$\log m \leq 2X, \qquad m := \prod_{p \in \mathcal{L}} p. \tag{5}$$

In particular, $\log N = o(L)$. If only positive upper relative density is assumed, the same conclusions hold along an unbounded sequence of X.

Proof. Choose $\delta > 0$ so that $|\mathcal{P} \cap [1, X]| \geq \delta X / \log X$ for large X. Proposition 2.1, with parameter $\delta/2$, removes at most $\delta X/(2 \log X)$ primes. The prime number theorem removes at most $\delta X/(4 \log X)$ further primes below $\delta X/8$. Take all remaining primes for $\mathcal{L}$. Every prime divisor of N is at most $X^{1-\eta}$, and its largest prime-power contribution is at most X. Thus $N \leq X^{X^{1-\eta}}$. Also $\log m \leq \sum_{p \leq X} \log p \leq 2X$ for large X. Finally,

$$\frac{\log N}{L} \leq \frac{4}{\delta} X^{-\eta} (\log X)^2 \longrightarrow 0.$$

For positive upper density, choose the initial sequence on which the first counting inequality holds and repeat the argument. □

Remark 2.3. The separate estimates for L and $\log N$ are essential. A bound of the form $N < m^{\kappa}$ alone does not supply the comparison needed below. All prime sets used in the applications are fixed completely split Chebotarev sets with finitely many primes removed, so the lower-density version applies for every sufficiently large X.

3 A local deficit for a curve cover

We record exactly which standard geometric facts are used. A smooth projective curve over a number field has a smooth proper model after inverting finitely many primes. A fixed finite morphism and finitely many auxiliary curves and maps extend after further exclusions; geometric integrality and degree can be retained. A smooth projective geometrically integral curve of genus g over $\mathbb{F}_p$ satisfies

$$\left| \#C(\mathbb{F}_p) - (p+1) \right| \leq 2g\sqrt{p}. \tag{6}$$

We use these reduction facts and the Hasse–Weil bound as established inputs, not as new results. The following argument is also consistent with the general thin-set estimate in [2, Lemma 3.8(2)].

Lemma 3.1 (Single-cover deficit)**.** *Let C/K be a smooth projective geometrically integral curve and let $f : C \to \mathbb{P}^1_K$ have degree $d \geq 2$. There are a finite Galois extension $E/\mathbb{Q}$ containing K and a finite set of rational primes S such that, whenever $p \notin S$ splits completely in E, for every prime ideal $\mathfrak{q}$ of K above p,*

$$\left| f_{\mathfrak{q}}(C_{\mathfrak{q}}(\mathbb{F}_p)) \cap \mathbb{F}_p \right| \leq \left(1 - \frac{1}{2d}\right) p. \tag{7}$$

The maps in this formula have good reduction and degree d.

Proof. Over an algebraic closure of K, the fiber product $C \times_{\mathbb{P}^1} C$ contains the diagonal and an irreducible component D_0 not equal to the diagonal. Indeed, in characteristic zero the map is separable, and its generic fiber has d distinct points; its self-product contains ordered pairs with unequal entries. The component D_0 and its normalization $\widetilde{D}_0$ are defined over a finite extension of K. The normalization is a smooth projective geometrically integral curve.

Take a finite Galois extension $E/\mathbb{Q}$ containing this field and all conjugate data. Exclude finitely many rational primes so that the curves, morphisms, degrees, and geometric integrality have good reduction. The normalization map to $D_0 \subset C \times C$ is an isomorphism away from a fixed finite subset. The intersection of D_0 with the diagonal is finite, because they are distinct components. After further prime exclusions, the number of points removed for these reasons is uniformly bounded in each residue field.

If p splits completely in E, all these data reduce over $\mathbb{F}_p$ at every prime of E above p. The Hasse–Weil bound on $\widetilde{D}_0$ therefore gives

$$p + O_f(\sqrt{p})$$

distinct ordered pairs $(P, Q) \in C_{\mathfrak{q}}(\mathbb{F}_p)^2$ with $P \neq Q$ and $f(P) = f(Q)$. The error constant can be chosen uniformly among the finitely many conjugate reductions. The same bound gives $\#C_{\mathfrak{q}}(\mathbb{F}_p) = p + 1 + O_f(\sqrt{p})$.

For $u \in \mathbb{P}^1(\mathbb{F}_p)$ let r_u count the rational points of the fiber. Since the reduced map has degree d, $0 \leq r_u \leq d$. Consequently

$$\begin{aligned} \#C_{\mathfrak{q}}(\mathbb{F}_p) - |f_{\mathfrak{q}}(C_{\mathfrak{q}}(\mathbb{F}_p))| &= \sum_{r_u > 0} (r_u - 1) \\ &\geq \frac{1}{d} \sum_u r_u(r_u - 1) \\ &\geq \frac{p}{d} - O_f(\sqrt{p}). \end{aligned}$$

It follows that the projective image, and hence its finite part, has cardinality at most $(1 - 1/d)p + O_f(\sqrt{p})$. For all sufficiently large p this is at most $(1 - 1/(2d))p$. Absorb the remaining primes into S. □

Lemma 3.2 (Reduction of a represented integral value)**.** *At a good prime ideal $\mathfrak{q}$, suppose $a(z)$ is a unit, $h(z)$ is integral, and*

$$t = a(z)f(P) + h(z) \in \mathcal{O}_K, \qquad P \in C(K), \quad f(P) \in K.$$

Then the reduction of t lies in

$$\overline{a(z)}\,(f_{\mathfrak{q}}(C_{\mathfrak{q}}(\mathbb{F}_p)) \cap \mathbb{F}_p) + \overline{h(z)}.$$

Proof. The identity $f(P) = (t - h(z))/a(z)$ makes $f(P)$ integral at $\mathfrak{q}$. Properness extends P to a point of the proper model over the valuation ring $\mathcal{O}_{K,\mathfrak{q}}$. The extended morphism commutes with specialization. Since $f(P)$ is integral, its projective specialization is finite and is exactly its residue in $\mathbb{F}_p$. Reducing the displayed affine identity proves the assertion. □

No affine coordinate of P has been assumed integral. This matters even when $C = \mathbb{P}^1$: a rational point can have a denominator, and the map can have poles. Projective good reduction and properness handle both issues.

4 Compression and the Chinese remainder theorem

Choose a finite presentation of T as in Definition 1.1. Treat each point of its finite part as a separate piece. Enlarge one finite Galois extension $E/\mathbb{Q}$ to contain K and the extensions needed in Lemma 3.1 for every cover. There are finitely many pieces, say J; the empty case is immediate and may be discarded.

Choose a finite set S of rational primes with the following properties. Outside primes above S, every coordinate of every generator of Γ is a unit; c in $a(z) = cz^{\boldsymbol{\alpha}}$ is a unit; all coefficients of h and all finite exceptional values are integral; and every curve and map used above has good reduction. Since only finitely many elements of K are involved, such S exists. It follows that these unit and integrality assertions hold for *every* $z \in \Gamma$, including negative powers of generators.

Let $\mathcal{P}$ consist of rational primes outside S that split completely in E. Chebotarev gives positive density, and

$$\mathcal{O}_K/p\mathcal{O}_K \simeq \prod_{\mathfrak{q}\mid p} \mathbb{F}_p \simeq \mathbb{F}_p^D \qquad (p \in \mathcal{P}). \tag{8}$$

After removing finitely many additional primes, there is one $\theta \in (0,1)$ such that each piece has at most θp possible finite residues at each $\mathfrak{q} \mid p$. A singleton has at most one residue, so it satisfies the same assertion for large p.

Proposition 4.1 (Finite-modulus bound)**.** *For any finite $\mathcal{L} \subset \mathcal{P}$, with $m = \prod_{p\in\mathcal{L}} p$, $N = \operatorname{lcm}_{p\in\mathcal{L}}(p-1)$, and $L = |\mathcal{L}|$,*

$$\frac{|\mathcal{R} \bmod m\mathcal{O}_K|}{m^D} \le JwN^r\theta^{DL}, \tag{9}$$

$$\frac{|(\mathcal{R}\cap\mathbb{Z}) \bmod m\mathbb{Z}|}{m} \le JwN^r\theta^{L}. \tag{10}$$

Proof. For each $p \in \mathcal{L}$ and $\mathfrak{q} \mid p$, the multiplicative group $\mathbb{F}_p^\times$ has exponent $p-1$, which divides N. Therefore the simultaneous reduction of Γ at all these primes factors through Γ/Γ^N. Its size is at most wN^r by (1).

Fix *both* a piece i of the presentation of T and a class $\xi \in \Gamma/\Gamma^N$. At every $\mathfrak{q} \mid p \in \mathcal{L}$, the reductions of $a(z)$ and $h(z)$ are fixed for $z \in \xi$, and the former is nonzero. The affine map $y \mapsto \overline{a(z)}y + \overline{h(z)}$ is a bijection of $\mathbb{F}_p$. Lemma 3.2, or direct reduction for a singleton piece, therefore places the corresponding values of t in at most θp residue classes at that prime ideal.

The Chinese remainder theorem and (8) show that, for this fixed pair (i,ξ), there are at most

$$\prod_{p\in\mathcal{L}} (\theta p)^D = \theta^{DL} m^D$$

possible residue classes modulo $m\mathcal{O}_K$. Only now take the union over i and ξ. This proves (9); no independence of the parameter residues at distinct primes is needed.

For rational integers, choose one prime ideal $\mathfrak{q}_p$ over each $p \in \mathcal{L}$. The residue of an integer in $\mathcal{O}_K/\mathfrak{q}_p$ is its usual residue in $\mathbb{F}_p$. Repeating the argument with just these L coordinates gives at most $\theta^L m$ residues for a fixed (i,ξ) and proves (10) after taking the same union. □

Remark 4.2 (Why the finite union is taken last)*.* One cannot assume a local deficit for the union of all thin pieces. For example, the three thin sets defined by the existence of a rational solution to

$$y^2 = t, \qquad y^2 = t-1, \qquad y^2 = t(t-1)$$

cover all finite-field residues at every odd prime: if neither t nor $t-1$ is a square, their product is a square. Their union is not all of $\mathbb{Q}$, since $t=3$ gives the three nonsquares $3, 2, 6$. In Proposition 4.1, membership in one fixed global piece is imposed across every prime before the finite union is taken.

Proof of Theorem 1.2. Apply Lemma 2.2 to $\mathcal{P}$. Since $\log N = o(L)$, the logarithms of the right sides of (9) and (10) tend to $-\infty$. The first residue assertion follows. Each residue class modulo $m\mathcal{O}_K$ has density m^{-D} in the chosen boxes, so the same assertion already proves qualitative density zero.

For the quantitative conclusion, take

$$X = \tfrac{1}{4}\log H$$

for sufficiently large H. The compression construction gives $m \le e^{2X} \le H^{1/2}$. Write $\beta = -\log\theta > 0$. By (3)–(4),

$$\log(Jw) + r\log N = o(X/\log X), \qquad L \ge \frac{\delta X}{4\log X}.$$

Thus, for all large H, the residue proportions in (9) and (10) are respectively bounded by

$$\exp\{-D\beta\delta X/(8\log X)\}, \qquad \exp\{-\beta\delta X/(8\log X)\}.$$

Each residue class modulo $m\mathcal{O}_K$ contains at most $(2H/m+2)^D$ elements of $\mathcal{B}_K(H)$. Multiplying this by the number of admissible classes bounds the exceptional count by

$$(2H+2m)^D \exp\{-D\beta\delta X/(8\log X)\}.$$

Since $m \le H^{1/2}$ and $X = \frac{1}{4}\log H$, this has the required form. For the rational-integer slice, each class modulo m contains at most $2H/m+2$ integers in $[-H, H]$. Use (10) in exactly this one-dimensional count. This proves (iii) separately. □

5 Simultaneous irreducibility and examples

We spell out the standard thin-set specialization argument, including full Galois groups. The specialization of a finite Galois extension at an unramified parameter has group a subgroup of the generic group; quotient covers detect its fixed cosets. These are classical facts of specialization theory; see [3, Proposition 3.3.1].

Lemma 5.1 (Thin bad-specialization set)**.** *For the finite list P_i in Theorem 1.3, the parameters at which any of the stated degree, separability, irreducibility, or Galois group properties fails are contained in a thin subset of K.*

Proof. Fix i and let $L_i/K(T)$ be its splitting field. Remove the finite set of poles, leading-coefficient zeros, discriminant zeros, and branch values. At another parameter $y \in K$, its specialized Galois group $G_{i,y}$ is a subgroup of G_i. If it is proper, it is contained in a conjugate of a maximal proper subgroup $H < G_i$. The specialized action on G_i/H then fixes a coset. Equivalently, y lifts to a K-point on the smooth projective quotient curve with function field L_i^H. Its map to $\mathbb{P}^1$ has degree $[G_i : H] \ge 2$.

There are finitely many such subgroups. A smooth integral quotient curve that is not geometrically integral has a constant field strictly larger than K and has no K-point; it contributes no values. The geometrically integral quotient curves give pieces of Definition 1.1,

and the finitely many removed values give its finite part. Thus parameters with a proper specialized group form a thin set. With the full generic group, its action remains transitive because P_i is generically irreducible, so the specialized polynomial is irreducible as well. Take the finite union over i. □

Proof of Theorem 1.3. Let T be the thin bad set from Lemma 5.1. A failing t has, for some $z \in \Gamma$,

$$y = a(z)t + h(z) \in T, \qquad t = a(z)^{-1}y - a(z)^{-1}h(z).$$

The reciprocal a^{-1} is a nonzero monomial and $-a^{-1}h$ is a Laurent polynomial. Theorem 1.2, applied to these two functions, bounds the set of all failing t and proves both the ambient and rational-integer conclusions. □

The inverse-affine formulation $P_i(a(z)^{-1}(t-h(z)), Y)$ is equivalent, by the same substitution. One may also preserve the full generic Galois group of the compositum of all splitting fields: apply the maximal subgroup argument to that one extension. This retains generic joint Galois relations, not merely the individual groups.

Universal Hilbert subsets have a substantial earlier literature. Dèbes–Zannier [7] construct explicit universal sequences, and Zannier's overview [8, Theorem H] describes further power-sum results. Those statements fix a sequence and, for each fixed polynomial, allow finitely many bad sequence members. Here the polynomials and multiplicative affine family are fixed, and a density-one set of translations is good for *every* member of that family. This quantifier distinction is useful, but is not a claim to the first universal Hilbert theorem. In particular, no one parameter is asserted to work for all polynomials chosen afterwards.

Corollary 5.2 (Fixed numbers of multiplicative translates)**.** *Let $G \leq K^\times$ be finitely generated, let $T \subset K$ be thin, and let $b_1, \dots, b_s \in K$ be fixed. Then*

$$\mathcal{O}_K \cap (T + b_1G + \cdots + b_sG)$$

satisfies the three conclusions of Theorem 1.2. The same is true with T replaced by GT.

Proof. Use $\Gamma = G^s$, $a = 1$, and $h(z) = \sum b_j z_j$. For the final assertion use $\Gamma = G^{s+1}$ and $a(z) = z_0$. □

Example 5.3 (Odd powers and signed shifts)**.** For every $d \geq 2$, the image $\{x^d : x \in K\}$ is thin. For fixed finitely many b_j and G, almost every algebraic integer t admits no representation

$$t + b_1u_1 + \cdots + b_su_s = x^d, \qquad x \in K, \quad u_j \in G.$$

The same density-one statement holds among rational integers, even when $K \neq \mathbb{Q}$. In particular, $d = 3$ is included. Negative exponents in the u_j are unrestricted, so a positive-term counting argument would not be an adequate substitute for the congruence proof.

Example 5.4 (A nonpolynomial rational map)**.** The map $f(x) = x + x^{-1}$ has degree two as a map $\mathbb{P}^1 \to \mathbb{P}^1$. Its finite values form a thin set. Thus almost every t avoids

$$t + b_1u_1 + \cdots + b_su_s = x + x^{-1}, \qquad x \in K^\times, \quad u_j \in G,$$

with the quantitative exceptional bounds above. Points at a pole are excluded by the requirement that the displayed value be finite.

Example 5.5 (A cover from a nonrational curve)**.** Let $C/\mathbb{Q}$ be the smooth projective completion of

$$v^2 = u^3 - 2.$$

It is a genus-one curve, hence not a rational curve. The coordinate u defines a degree-two morphism $f : C \to \mathbb{P}^1$. Its finite rational image is thin. Consequently, for fixed rational coefficients b_j and a fixed finitely generated $G \leq \mathbb{Q}^\times$, the integers of the form

$$u + b_1 u_1 + \cdots + b_s u_s, \qquad (u, v) \in C(\mathbb{Q}), \quad u_j \in G,$$

have the asserted density-zero bound. The theorem applies whether this particular curve has finitely or infinitely many rational points; no rank computation is needed. This example illustrates that the proof is not restricted to rational parametrizations.

Example 5.6 (Uniform irreducibility)**.** The polynomial $Y^3 - Y - T$ is irreducible in $\mathbb{Q}(T)[Y]$: the extension $\mathbb{Q}(Y)/\mathbb{Q}(Y^3 - Y)$ has degree three. Fix a finitely generated $G \leq \mathbb{Q}^\times$ and rational $b_1, \ldots, b_s$. Outside a quantitatively sparse set of integers t, the polynomial

$$Y^3 - Y - (u_0 t + b_1 u_1 + \cdots + b_s u_s)$$

is irreducible over $\mathbb{Q}$ for every $(u_0, \ldots, u_s) \in G^{s+1}$. The conclusion can be required simultaneously for any fixed finite list of irreducible families.

6 Polynomial families with varying coefficients

Theorem 1.4 allows the lower coefficients to vary with z, so the fixed-cover proof cannot simply be reused. A direct finite-field argument gives the needed uniform gap.

Lemma 6.1 (Elementary value-set bound)**.** *Let p be prime, let $d \geq 2$ divide $p - 1$, and let $G \in \mathbb{F}_p[X]$ have degree exactly d. Then*

$$|G(\mathbb{F}_p)| \leq p - \frac{p-1}{d}.$$

Proof. Put $k = (p-1)/d$ and let $a \neq 0$ be the leading coefficient. The sums of powers in $\mathbb{F}_p$ imply

$$S_j := \sum_{x \in \mathbb{F}_p} G(x)^j = 0 \quad (1 \leq j < k), \qquad S_k = -a^k \neq 0.$$

Indeed, below degree $p-1$ every nonconstant monomial has zero sum, as does the constant monomial in characteristic p; at degree $p-1$ only the leading term contributes.

Set $Q(T) = \prod_{x \in \mathbb{F}_p} (T - G(x))$. Since $k < p$, Newton's identities with invertible indices give

$$Q(T) = T^p + O(T^{p-k}),$$

where the notation means that all remaining terms have degree at most $p-k$. Let $V = G(\mathbb{F}_p)$, let $m = p - |V|$, and let r_v be the multiplicity of v in this list of values. Define monic degree-m polynomials

$$A(T) = \prod_{v \in V} (T - v)^{r_v - 1}, \qquad B(T) = \prod_{b \in \mathbb{F}_p \setminus V} (T - b).$$

They satisfy

$$Q(T)B(T) = (T^p - T)A(T). \tag{11}$$

If $m < k$, then $p - k + m \leq p - 1$ and $m + 1 < p$. The terms of degrees at least p on the left of (11) are exactly $T^p B(T)$; on the right they are exactly $T^p A(T)$. Thus $A = B$, and cancellation gives $Q = T^p - T$. This would make G a permutation of $\mathbb{F}_p$. But then $S_k = \sum_{y \in \mathbb{F}_p} y^k = 0$, because $1 \leq k < p - 1$, a contradiction. Therefore $m \geq k$. □

Remark 6.2. The lemma is a special finite-field value-set estimate, not a new classification of exceptional polynomials. The displayed proof is included so that neither a parity assumption nor such a classification is needed in the application.

Proof of Theorem 1.4. Choose a finite exceptional set S so that, outside primes above S, all generator coordinates are units, c is a unit, and every coefficient of every A_i is integral. Take a finite Galois extension $E/\mathbb{Q}$ containing K and $\mathbb{Q}(\zeta_d)$. The rational primes outside S splitting completely in E have positive density, split completely in K, and satisfy $p \equiv 1 \pmod{d}$.

At a prime ideal $\mathfrak{q}$ over one of these primes, fix $z \in \Gamma$. The specialized polynomial $F(X, z)$ has integral coefficients and unit leading coefficient. If $v_{\mathfrak{q}}(x) < 0$, its leading term has valuation $dv_{\mathfrak{q}}(x)$, strictly smaller than the valuation of every lower-degree term: for $i < d$,

$$v_{\mathfrak{q}}(A_i(z)x^i) \geq iv_{\mathfrak{q}}(x) > dv_{\mathfrak{q}}(x).$$

A sum with a unique term of least valuation has that valuation. Therefore $F(x, z)$ could not be integral. Every represented integral value thus has x integral at $\mathfrak{q}$.

The reduction of $F(X, z)$ has degree exactly d. Lemma 6.1 gives a relative value-set bound

$$1 - \frac{p-1}{dp} \leq \theta := 1 - \frac{1}{2d} < 1.$$

For any compressed set of primes, fix a class $\xi \in \Gamma/\Gamma^N$. Its reductions fix all the coefficients of $F(X, z)$ simultaneously, so the bound holds for that one polynomial at each prime ideal. The same CRT calculation as Proposition 4.1, now with $J = 1$, yields

$$\frac{|\mathcal{R}_F \bmod m\mathcal{O}_K|}{m^D} \leq wN^r\theta^{DL}.$$

Selecting one prime ideal above each rational prime similarly gives $wN^r\theta^L$ for the rational-integer slice. Apply Lemma 2.2 and the box count from the proof of Theorem 1.2. All three assertions follow. □

Example 6.3 (A genuinely parameter-dependent odd-degree family)**.** Let $\Gamma \leq (K^\times)^2$ be finitely generated. The integral values of

$$F(X, z_1, z_2) = z_1X^3 + (z_2 + z_2^{-1})X^2 + (1 + z_1^{-1})X + z_2$$

with $X \in K$ and $(z_1, z_2) \in \Gamma$ form a set of density zero, with both quantitative bounds. The quadratic and linear coefficients vary, so this assertion is not obtained by identifying one fixed image $f(C(K))$ and translating it.

7 Fixed degrees and unrestricted proper powers

The finite-union principle allows any fixed finite collection of proper-power degrees. It does not allow all $d \geq 2$ simultaneously. This is not just a matter of losing a uniform constant in the proof.

Proposition 7.1 (Squarefree-modulus obstruction)**.** *Let $U = \{x^d : x \in \mathbb{Q},\ d \geq 2\}$. For every squarefree positive integer M, the integer proper powers, and hence $U \cap \mathbb{Z}$, map surjectively onto $\mathbb{Z}/M\mathbb{Z}$.*

Proof. For $M = 1$ the assertion is immediate. Otherwise choose a prime q not dividing $\operatorname{lcm}_{p|M}(p-1)$. The map $x \mapsto x^q$ is a bijection of each $\mathbb{F}_p^\times$ and fixes zero, hence is a bijection of $\mathbb{F}_p$. The Chinese remainder theorem makes it a bijection of $\mathbb{Z}/M\mathbb{Z}$. For each residue, an integer representative of its preimage has an integer qth power in that residue. □

Thus the arbitrarily small squarefree-residue conclusion of Theorem 1.2 is false for U itself. A countable union of fixed-degree conclusions cannot prove it. Although integer proper powers without shifts have density zero by elementary counting, the present theorem does not prove the corresponding assertion after all allowed S-unit shifts. Nor can the exponent be bounded merely from the target size: the identity

$$-1 + 2^q + 3^0 = 2^q$$

gives a fixed target with representations of unbounded prime degree. An all-degree tail estimate would require an additional argument that handles such cancellation. None is claimed here.

8 Scope and remaining limitations

The main conclusion is a transfer theorem: thin images and suitable polynomial value sets remain sparse among integral targets after all allowed multiplicative affine operations. It does not classify the solutions for a fixed target, prove that the resulting set is thin, or settle a general local–global principle.

The monomial assumption has a specific role. It makes $a(z)$, or the leading coefficient of $F(X, z)$, a unit at every chosen prime for every $z \in \Gamma$. A general Laurent polynomial can vanish modulo selected primes even if it never vanishes on Γ in characteristic zero. The proof does not justify replacing the monomial by an arbitrary nonvanishing function.

Likewise, a general parameter-dependent rational map $F(X, z)$ need not have one fixed extension over which all its relevant off-diagonal components are defined. The fixed Chebotarev set in the proof may then be unavailable. No theorem for all such rational families is claimed. The number of multiplicative summands is fixed, and no assertion is made for an unbounded number of summands. The quantitative saving is subpower, not a fixed power saving.

The square-value question stated by Corvaja and Zannier concerns whether many integers t avoid all squares $t + a2^m + b3^n$, with $m, n \in \mathbb{Z}$, and analogous S-unit shifts [4, after Example 1.3]. It is a legitimate earlier question, and the square specialization here answers it. However, that conclusion already occurs in [5]; it is not a new resolution counted in this paper. The extensions to arbitrary thin covers, simultaneous affine specialization, the separate rational-integer slice, and all-degree number-field polynomial families are the results proved here. Their historical priority has not been established by an exhaustive literature search.

This manuscript does not establish a new complete solution of a previously named open problem under all the stated novelty requirements. It does not resolve Baker's question for arbitrary moduli. Appendix A proves only its displayed family by an independent method. No assertion is made on the rational-base Lambert conjecture at $3/2$ or on Mills-type constants. The proofs are unconditional relative to the explicitly identified classical theorems, but have not been certified in a proof assistant or subjected to external peer review.

A An independent application to Baker's question

Baker asked whether, for every integer $q \geq 3$, the numbers $L(1,\chi)$ as χ ranges over the nonprincipal Dirichlet characters modulo q are linearly independent over $\mathbb{Q}$ [9, p. 48]; see [12] for the history and the precise parity reduction. The coprimality case $\gcd(q,\varphi(q)) = 1$ follows from Baker–Birch–Wirsing [10]. The argument below is independent of the thin-set construction in the main text. It proves an unconditional infinite family by a signed Galois interchange, not by a general statement about arbitrary twisted cotangent sums. It does not resolve the question for all moduli or all prime powers, and no claim of bibliographic priority is made.

Write $\lambda(n)$ for the exponent of $(\mathbb{Z}/n\mathbb{Z})^\times$, with $\lambda(1) = 1$.

Theorem A.1. *Let $m \geq 1$ be odd and suppose that*

$$\gcd(m, \lambda(m)) = 1, \qquad 8 \nmid \lambda(m).$$

Then the numbers $L(1,\chi)$, with χ ranging over all nonprincipal Dirichlet characters modulo $32m$, are linearly independent over $\mathbb{Q}$. In particular, this holds for $q = 32r$ whenever r is an odd prime satisfying $r \not\equiv 1 \pmod 8$.

The final assertion supplies infinitely many moduli, since the primes $r \equiv 3 \pmod 8$ alone form an infinite set. For these moduli $\gcd(q, \lambda(q)) = 8$; thus they are not covered by the elementary singleton-fibre criterion proved below.

A.1 The external parity input

We use two established facts for a fixed modulus $q > 2$:

1. The values $L(1,\chi)$ at nonprincipal even characters are linearly independent over $\overline{\mathbb{Q}}$, by Murty–Murty [11]; the statement is also recorded explicitly in [12, Theorem 3].
2. The $\mathbb{Q}$-span of these even values has zero intersection with the $\mathbb{Q}$-span of the odd values [12, Theorem 5].

The theorem numbering here refers to the cited arXiv version of [12]. For the second fact, the underlying input is Baker's theorem applied to logarithms of positive real algebraic numbers together with $\log(-1) = \pi i$; see also the logarithm lemma of Murty–Saradha [13], reproduced as Lemma 4 in [12]. An even value is an algebraic linear combination of such real logarithms, whereas an odd value is an algebraic multiple of π. Consequently it suffices to prove rational linear independence of the odd values. The restriction-fibre framework is developed in the earlier corpus manuscript [6] and is reproved below for completeness. The additional step here is the signed interchange within the two-element fibres and its application to the displayed modulus family.

A.2 Algebraic representatives and restriction fibres

For $q \geq 3$, put

$$U_q = (\mathbb{Z}/q\mathbb{Z})^\times, \qquad \lambda = \lambda(q), \qquad N = \operatorname{lcm}(q,\lambda), \qquad \zeta_n = e^{2\pi i/n}.$$

For $a \in U_q$ and an odd character χ modulo q, define

$$y_a = \frac{\zeta_q^a + 1}{\zeta_q^a - 1} = -i \cot \frac{\pi a}{q}, \qquad W_\chi = \sum_{a \in U_q} \chi(a) y_a \in \mathbb{Q}(\zeta_N).$$

The partial-fraction expansion of the cotangent gives

$$L(1,\chi) = \frac{\pi}{2q} \sum_{a \in U_q} \chi(a) \cot \frac{\pi a}{q} = \frac{\pi i}{2q} W_\chi. \tag{12}$$

Indeed, oddness makes the symmetric sum of $\chi(n)/n$ over nonzero integers equal to twice the ordinary Dirichlet series; grouping the symmetric sum by residue classes and using $\sum_{k \in \mathbb{Z}} (a + kq)^{-1} = (\pi/q) \cot(\pi a/q)$ proves (12). All sums over k here are symmetric limits. Dirichlet nonvanishing implies $W_\chi \neq 0$.

For $u \in U_N$, let σ_u denote the automorphism of $\mathbb{Q}(\zeta_N)$ sending ζ_N to ζ_N^u. Since character values have orders dividing λ, a change of variable $b = ua$ gives

$$\sigma_u(W_\chi) = \sum_{a \in U_q} \chi(a)^u y_{ua} = \overline{\chi^u}(u) W_{\chi^u}. \tag{13}$$

The exponent u on a character and its argument u are interpreted modulo the character order and modulo q, respectively. Only the algebraic numbers W_χ are subjected to Galois automorphisms; no action on the transcendental common factor in (12) is asserted.

Lemma A.2 (Restriction-fibre splitting)**.** *Put*

$$\delta = \gcd(q, \lambda), \qquad K = \mathbb{Q}(\zeta_\lambda), \qquad H = \ker(U_q \longrightarrow U_\delta).$$

Every rational linear relation among the odd numbers W_χ splits into separate relations among characters having the same restriction to H. Each nonempty fibre is a coset of the group of even characters modulo δ, viewed as characters modulo q. In particular, if $\delta = 4$ all odd W_χ are linearly independent over $\mathbb{Q}$. If $\delta = 8$, every odd fibre is a pair $\{\chi, \chi\eta_8\}$, where

$$\eta_8(a) = (-1)^{(a^2-1)/8} \qquad (a \text{ odd})$$

is the nonprincipal even character modulo 8.

Proof. The reduction map $U_q \longrightarrow U_\delta$ is onto: a unit modulo a divisor of q lifts to a unit modulo q by the Chinese remainder theorem at the prime-power factors. For each $s \in H$, the congruences

$$u_s \equiv s \pmod{q}, \qquad u_s \equiv 1 \pmod{\lambda}$$

have a solution, because $s \equiv 1 \pmod{\delta}$. The solution is a unit modulo N and is unique modulo N. These automorphisms identify H with $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/K)$. Equation (13) then reads

$$\sigma_{u_s}(W_\chi) = \overline{\chi}(s) W_\chi.$$

All characters of H have values in K, since the exponent of H divides λ. For a character μ of H, the K-linear operator

$$E_\mu = \frac{1}{|H|} \sum_{s \in H} \mu(s) \sigma_{u_s}$$

sends W_χ to W_χ if $\chi|_H = \mu$, and to zero otherwise, by character orthogonality. Applying E_μ to a rational relation proves the splitting assertion.

Two characters agree on H exactly when their quotient factors through $U_q/H = U_\delta$. If both are odd, their quotient is even; conversely multiplying by an even character modulo δ preserves both oddness and the restriction to H. Modulo 4 there is no nonprincipal even character, so every fibre is a singleton and nonvanishing proves independence. Modulo 8 the even character group is $\{1, \eta_8\}$, which proves the last assertion. □

A.3 A signed Galois interchange

Lemma A.3. *Let η be an even quadratic character modulo q, and let χ be odd. Suppose that a unit $u \in U_N$ satisfies*

$$\chi^u = \chi\eta, \qquad \eta(u) = -1.$$

Then W_χ and $W_{\chi\eta}$ are linearly independent over $\mathbb{Q}$. In particular, $L(1,\chi\eta)/L(1,\chi)$ is algebraic and irrational.

Proof. All these values are nonzero. A rational dependence would therefore give

$$c = \frac{W_{\chi\eta}}{W_\chi} \in \mathbb{Q}^\times.$$

The integer u is odd because λ is even, and hence $\eta^u = \eta$. Thus $(\chi\eta)^u = \chi$. By (13),

$$\sigma_u(c) = \frac{\overline{\chi}(u)W_\chi}{\overline{\chi\eta}(u)W_{\chi\eta}} = -\frac{1}{c}.$$

Rationality would give $c = \sigma_u(c)$ and then $c^2 = -1$, a contradiction. Formula (12) identifies the two ratios and also proves that the ratio of the L-values is algebraic. □

Proof of Theorem A.1. Set $\ell = \lambda(m)$ and $q = 32m$. The hypotheses imply

$$\lambda(q) = \operatorname{lcm}(8,\ell), \qquad \gcd(q,\lambda(q)) = 8.$$

Indeed the odd part of the gcd is trivial because $\gcd(m,\ell) = 1$, while the 2-part of $\lambda(q)$ is exactly 8 because $8 \nmid \ell$. Lemma A.2 therefore reduces the odd-value assertion to the independence of each pair $\{W_\chi, W_{\chi\eta_8}\}$.

By the Chinese remainder theorem, write $\chi = \alpha\beta$ with α modulo 32 and β modulo m. The convention for $m = 1$ is the trivial character. Since

$$U_{32} = \langle -1\rangle \times \langle 5\rangle, \qquad \operatorname{ord}_{32}(5) = 8,$$

we may write $\alpha(5) = \zeta_8^j$ for $0 \le j < 8$.

First suppose that j is odd. Choose an integer u satisfying

$$u \equiv 5 \pmod{32}, \qquad u \equiv 1 \pmod{m\ell}. \tag{14}$$

These congruences are compatible because

$$\gcd(32, m\ell) = \gcd(32,\ell) \in \{1,2,4\},$$

and this gcd divides $5-1$. Every prime factor of $N = \operatorname{lcm}(q,\lambda(q))$ divides $32m\ell$, so the residues in (14) make u a unit modulo N. The second congruence fixes the values of β and hence $\beta^u = \beta$. The first gives $u \equiv 5 \pmod 8$, so

$$\alpha^u = \alpha^5 = \alpha\eta_8.$$

For the last identity it suffices to check the generators: $\alpha^4(-1) = 1$ and $\alpha^4(5) = \zeta_8^{4j} = -1 = \eta_8(5)$. Thus $\chi^u = \chi\eta_8$ and $\eta_8(u) = \eta_8(5) = -1$. Lemma A.3 proves independence of the pair.

Now suppose that j is even. The kernel of $U_{32} \longrightarrow U_{16}$ is $\{1,17\}$, and $17 = 5^4 \pmod{32}$. Thus

$$\alpha(17) = \zeta_8^{4j} = 1,$$

so α factors modulo 16. The same is true of $\alpha\eta_8$, whose exponent at 5 is $j+4$. The two distinct odd characters in the pair therefore descend to distinct odd characters modulo $v = 16m$. Their arithmetic functions on the integers are unchanged: the two moduli $32m$ and $16m$ have the same prime divisors, so both the values on units and the zero values on nonunits agree. Their Dirichlet L-values are consequently unchanged, without an additional Euler factor. But

$$\lambda(v) = \operatorname{lcm}(4, \ell), \qquad \gcd(v, \lambda(v)) = 4,$$

so the singleton-fibre case of Lemma A.2 proves independence of all odd values modulo v, and in particular of this pair.

Both cases are now settled. Restriction-fibre splitting proves independence of all odd values modulo $32m$. The two established parity facts stated above then imply independence of all nonprincipal values. Finally, for an odd prime r, $\lambda(r) = r-1$ and $\gcd(r, r-1) = 1$, so the displayed prime family follows immediately. □

Theorem A.4 (Transcendence consequence)**.** *For a modulus $q = 32m$ satisfying the hypotheses of Theorem A.1, every rational linear combination*

$$\sum_{\substack{\chi \bmod q \\ \chi \neq \chi_0}} c_\chi L(1, \chi), \qquad c_\chi \in \mathbb{Q},$$

whose coefficients are not all zero is transcendental.

Proof. Theorem A.1 makes the displayed value nonzero. For completeness, if f is a periodic algebraic-valued function of period q with mean zero, Fourier inversion and Abel summation give

$$L(1, f) = -\sum_{a=1}^{q-1} \widehat{f}(a) \log(1 - \zeta_q^a), \qquad \widehat{f}(a) = \frac{1}{q} \sum_{b=1}^{q} f(b) \zeta_q^{-ab},$$

with the logarithms obtained by radial continuation from the unit disc. Apply this to $f = \sum_\chi c_\chi \chi$. All coefficients and logarithm arguments are algebraic, and all arguments are nonzero. Baker's theorem on linear forms in logarithms [9] says that such a linear form, if nonzero, is transcendental. This proves the assertion. □

The individual transcendence of the $L(1, \chi)$ is classical. The consequence here is simultaneous nonvanishing, and hence transcendence, for every nontrivial rational combination in the stated family. It does not assert independence of the odd values over $\overline{\mathbb{Q}}$; those values are all algebraic multiples of π.

Scope. The proof gives the full nonprincipal-character assertion for every modulus in Theorem A.1, rather than a rank estimate or a finite verification. It also proves that the ratios for pairs with primitive 32-part are algebraic and irrational. It makes essential use of two-element restriction fibres and the sign $\eta_8(u) = -1$. Neither the proof nor the theorem asserts the corresponding result for arbitrary moduli or for every power of 2.

B Reproducible finite illustrations

The proofs above do not depend on a computation. The following standalone Python 3 program illustrates Lemma 6.1 by exhausting 377 304 polynomials over the listed small fields, and checks the finite-union warning of Remark 4.2 at every odd prime below 100. It uses only the standard library. The expected output is `PASS: 377304 polynomials; 24 odd primes`. These finite tests do not prove the statements for arbitrary fields or arbitrary parameters, and are not proof-assistant certificates.

```
from itertools import product
from math import isqrt


def is_prime(n):
    return n > 1 and all(n % d for d in range(2, isqrt(n) + 1))


count = 0
for p in (3, 5, 7, 11, 13):
    for d in range(2, 5):
        if (p - 1) % d:
            continue
        bound = p - (p - 1) // d
        powers = [[pow(x, j, p) for j in range(d + 1)]
                  for x in range(p)]
        for low in product(range(p), repeat=d):
            for lead in range(1, p):
                coefficients = low + (lead,)
                image = {
                    sum(c * v for c, v in zip(coefficients, row)) % p
                    for row in powers
                }
                assert len(image) <= bound, (p, d, coefficients)
                count += 1

prime_count = 0
for p in range(3, 100):
    if not is_prime(p):
        continue
    squares = {x * x % p for x in range(p)}
    for t in range(p):
        assert (t in squares or (t - 1) % p in squares
                or t * (t - 1) % p in squares), (p, t)
    prime_count += 1

assert count == 377304 and prime_count == 24
print(f"PASS: {count} polynomials; {prime_count} odd primes")
```

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